WORKED SOLUTION

STEP III 1987, Question 12

Mechanics • Variable mass

STEP III 1987, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q12

The key idea

This is a variable-mass problem, so $F = ma$ (or its rotational cousin) can't be used blindly. The safe method is to go back to first principles: compare the angular momentum about the pivot at time t with that at time $t + \delta t$, when a little burnt fuel has left the system. The pivot is smooth and gravity is vertical, so nothing exerts a moment about the vertical axis through the pivot, and that angular momentum is conserved.

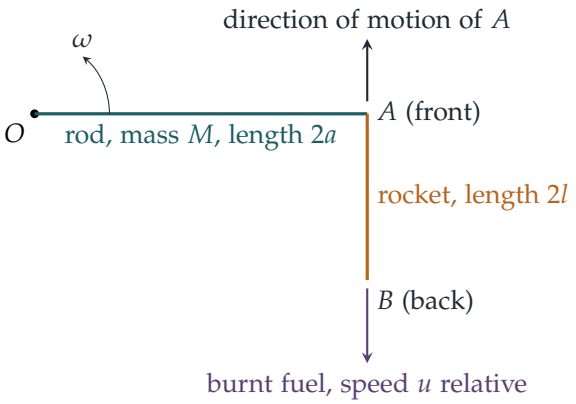
Solution

Setting up

Model the rod and the rocket as uniform rigid rods, rotating in a horizontal plane about the vertical axis through the pivot O. Let the rod be OA, with $OA = 2a$. Use unit vectors $\mathbf{e}_r$ along OA and $\mathbf{e}_\theta$ perpendicular to it in the direction of rotation, and let $\mathbf{k}$ be the upward vertical, so $\mathbf{e}_r \times \mathbf{e}_\theta = \mathbf{k}$.

The rocket's front end is at A and it points forwards, in the direction of motion of A. So the rocket lies along $\mathbf{e}_\theta$, extending *backwards* from A to its back end B, where

$$\overrightarrow{OB} = 2a\,\mathbf{e}_r - 2l\,\mathbf{e}_\theta.$$



Moment of inertia

The rod about its end has moment of inertia $\frac{1}{3}M(2a)^2 = \frac{4}{3}Ma^2$.

The rocket's centre is at $2a\,\mathbf{e}_r - l\,\mathbf{e}_\theta$, at distance $\sqrt{4a^2+l^2}$ from O. Its moment of inertia about its centre is $\frac{1}{3}ml^2$, so by the parallel axis theorem it contributes $\frac{1}{3}ml^2 + m(4a^2+l^2)$. The total is

$$I = \tfrac{4}{3}Ma^2 + m\left(4a^2 + \tfrac{4}{3}l^2\right).$$

How I changes

Since $m = 2\alpha l$ and $\dfrac{\mathrm{d}m}{\mathrm{d}t} = -\alpha\beta$, we have $\dfrac{\mathrm{d}l}{\mathrm{d}t} = -\frac{1}{2}\beta$. Differentiating I with the product rule,

$$\begin{aligned}\frac{\mathrm{d}I}{\mathrm{d}t} &= \frac{\mathrm{d}m}{\mathrm{d}t}\left(4a^2 + \tfrac{4}{3}l^2\right) + m\cdot\tfrac{8}{3}l\,\frac{\mathrm{d}l}{\mathrm{d}t}\\ &= -\alpha\beta\left(4a^2 + \tfrac{4}{3}l^2\right) + 2\alpha l\cdot\tfrac{8}{3}l\cdot\left(-\tfrac{1}{2}\beta\right) = -\alpha\beta\,(4a^2+4l^2).\end{aligned}$$

Notice that $4a^2 + 4l^2 = OB^2$: the moment of inertia drops at exactly the rate you'd get by removing mass from the back end B. That makes sense, because a uniform rocket that stays uniform, with its front fixed at A, can only shrink from the back.

Angular momentum of the ejected fuel

In the short time from t to $t+\delta t$ a mass $\delta\mu = \alpha\beta\,\delta t$ of burnt fuel leaves from B. Before it burnt it moved with the rocket, at velocity $\mathbf{v}_B = \omega\mathbf{k}\times\overrightarrow{OB}$. Afterwards its velocity is $\mathbf{v}_B - u\,\mathbf{e}_\theta$, since it moves directly backwards relative to the rocket. Its angular momentum about O is then

$$\begin{aligned}\delta\mu\,\overrightarrow{OB}\times(\mathbf{v}_B - u\,\mathbf{e}_\theta) &= \delta\mu\left[\omega\,OB^2\,\mathbf{k} - (2a\,\mathbf{e}_r - 2l\,\mathbf{e}_\theta)\times u\,\mathbf{e}_\theta\right]\\ &= \delta\mu\left[\omega(4a^2+4l^2) - 2au\right]\mathbf{k},\end{aligned}$$

using $\mathbf{e}_\theta\times\mathbf{e}_\theta = \mathbf{0}$ and $\mathbf{r}\times(\omega\mathbf{k}\times\mathbf{r}) = \omega r^2\mathbf{k}$ for $\mathbf{r}$ perpendicular to $\mathbf{k}$.

Ask yourself Why does only the $2a\,\mathbf{e}_r$ part of $\overrightarrow{OB}$ matter for the exhaust term?

The exhaust's relative velocity is along $\mathbf{e}_\theta$, the same line as the rocket. So the component of $\overrightarrow{OB}$ along the rocket has no moment. The perpendicular distance from O to the line of the exhaust (and of the thrust) is just $OA = 2a$, wherever along the rocket the fuel leaves.

Conservation of angular momentum

The pivot is smooth, so its reaction has no moment about the vertical axis through O, and gravity is vertical, so it has none either. The angular momentum about that axis of the whole system (firework plus the fuel burnt in δt) is therefore the same at t and $t + \delta t$:

$$I(t+\delta t)\,\omega(t+\delta t) + \delta\mu\left[\omega(4a^2+4l^2) - 2au\right] = I(t)\,\omega(t),$$

up to terms of order $(\delta t)^2$. Dividing by δt and letting $\delta t \to 0$, with $\delta\mu/\delta t = \alpha\beta$,

$$\frac{\mathrm{d}}{\mathrm{d}t}(I\omega) + \alpha\beta\,\omega(4a^2+4l^2) - 2\alpha\beta au = 0.$$

Expanding $\dfrac{\mathrm{d}}{\mathrm{d}t}(I\omega) = I\dfrac{\mathrm{d}\omega}{\mathrm{d}t} + \omega\dfrac{\mathrm{d}I}{\mathrm{d}t}$ and using $\dfrac{\mathrm{d}I}{\mathrm{d}t} = -\alpha\beta(4a^2+4l^2)$, the two ω terms cancel exactly, leaving

$$I\,\frac{\mathrm{d}\omega}{\mathrm{d}t} = 2\alpha\beta au.$$

This is just "moment of inertia $\times$ angular acceleration $=$ moment of the thrust": the thrust is $\alpha\beta u$, acting along the rocket at perpendicular distance $2a$ from O. Finally,

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = \frac{2\alpha\beta au}{\frac{4}{3}Ma^2 + 2\alpha l\left(4a^2 + \frac{4}{3}l^2\right)} = \frac{6\alpha\beta au}{4Ma^2 + 8\alpha l(3a^2+l^2)}$$
$$= \frac{3\alpha\beta au}{2\left[Ma^2 + 2\alpha l(3a^2+l^2)\right]},$$

as required. This holds while there is fuel left, that is, while $l > 0$.

Checks

- With no rocket ($l \to 0$) the result becomes $\frac{4}{3}Ma^2\,\dot{\omega} = 2a \cdot \alpha\beta u$: thrust times lever arm, as expected.
- Dimensions: $\alpha\beta u$ is a force (α is mass per length and β is a speed), so the numerator is force $\times$ length, and the denominator is a moment of inertia.
- Numerically, with $M = 1.3$, $a = 0.7$, $\alpha = 0.9$, $\beta = 0.4$, $u = 2.5$, $l(0) = 1.1$ and $\omega(0) = 0.3$, we stepped the motion by ejecting 2×10^5 small chunks from the back end and conserving angular momentum each time. At $t = 4.95$ this gave $\omega \approx 2.39817$, and integrating the formula gave 2.39816.

Common slips

- Writing $\dfrac{\mathrm{d}}{\mathrm{d}t}(I\omega) =$ torque for the firework alone. The firework isn't a closed system, and this gives the wrong answer unless you also account for the angular momentum carried off by the fuel.
- Giving the exhaust an absolute velocity of u backwards, rather than u relative to the moving rocket.
- Forgetting that l changes, so that the rocket's moment of inertia depends on time through both m and l.

Beyond STEP

The cancellation here is a general fact. For a body losing mass with the local rigid-body velocity plus a relative exhaust velocity, the equation of motion is the ordinary one for the instantaneous body, with an extra "thrust" force from the relative velocity. That is the rotational version of the rocket equation. Kleppner and Kolenkow's *An Introduction to Mechanics* treats variable-mass systems and angular momentum carefully, at just this level.

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